2. Work and Energy

1. Concept of Work

In common language, any activity is called work, but in physics, work has a specific meaning.
Work is said to be done only when a force applied to an object causes displacement of that object.

If you study for hours (mental work) or hold a heavy bag without moving, no *physical work* is done.

Mathematical Formula:

- Work (W) is the product of the magnitude of the force (F) and the displacement (s) in the direction of the force:
  $W = F \times s$

Work at an Angle:

- If the force is applied at an angle (θ) to the direction of displacement (like a child pulling a toy cart with a string), the formula becomes:
  $W = Fs \cos \theta$
- If $\theta = 0^\circ$ (same direction), work is maximum:
  $W = Fs$
- If $\theta = 90^\circ$ (perpendicular), work is zero
- If $\theta = 180^\circ$ (opposite direction), work is negative:
  $W = -Fs$

2. Units of Work

SI System:

- Unit of force: Newton (N)
- Unit of displacement: meter (m)
- Unit of work: **Joule (J)**

Definition of 1 Joule:
If a force of 1 Newton displaces an object through 1 meter in the direction of the force, the work done is 1 Joule.

CGS System:

- Unit of force: dyne
- Unit of displacement: centimeter (cm)
- Unit of work: **erg**

Relationship:

- **$1 \text{ Joule} = 10^7 \text{ erg}$**

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3. Positive, Negative, and Zero Work

Positive Work:

- When force and displacement are in the **same direction** ($\theta = 0^\circ$)
- Example: Pushing a stalled vehicle forward

Negative Work:

- When force and displacement are in **opposite directions** ($\theta = 180^\circ$)
- Example: Applying brakes to a moving car

Zero Work:

- When the applied force **does not cause displacement**, or
- When force and displacement are **perpendicular** ($\theta = 90^\circ$)

Example: An artificial satellite moving around the Earth in a circular orbit

4. Concept of Energy

Energy is the capacity of a body to perform work.

- Units of work and energy are the same:
    - SI unit: **Joule**
    - CGS unit: **erg**

Energy exists in various forms:

- Mechanical
- Heat
- Light
- Sound
- Chemical
- Nuclear
- Solar

5. Forms of Mechanical Energy

Mechanical energy is divided into two types:

A. Kinetic Energy (K.E.)

- The energy an object has because of its **motion**

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Formula:

- **$K.E. = \frac{1}{2} mv^2$**
  (where m is mass and v is velocity)
- The work done to displace a stationary object is the kinetic energy gained by that object.

B. Potential Energy (P.E.)

- The energy stored in an object because of its **specific state or position**

Formula:

- **$P.E. = mgh$**
  (where m is mass, g is gravitational acceleration, and h is height)
- **P.E. is relative:**
  The height of an object depends on the reference level; therefore, the P.E. will be different for different observers.

6. Law of Conservation of Energy

This law states that:

Energy can neither be created nor destroyed; it can only be converted from one form into another.

- The total amount of energy in the universe remains constant.

Examples of Energy Transformation:

- An electric lamp converts electrical energy into light energy
- An engine or fan converts electrical energy into mechanical energy
- Firecrackers convert chemical energy into light, sound, and heat energy

7. Free Fall and Total Energy

When an object falls solely under the influence of gravity, it is said to be in **free fall**.

- During free fall:
  - Potential Energy decreases
  - Kinetic Energy increases

Conservation Proof:

At any point during the fall (top, middle, or bottom):

- **Total Energy (T.E.) = K.E. + P.E.**
- Total Energy remains constant and equal to **mgh**

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8. Power

Power is the rate at which work is done.

- It measures how fast or slow work is performed.

Formula:

- **$Power (P) = \frac{Work (W)}{Time (t)}$**

Unit:

- SI unit of power: **Watt (W)**
- **1 Watt = 1 Joule/second**

2. Work and Energy

- Work and Energy
  - Work
    - Definition
      - Force applied causing displacement
    - Formula
      - $W = F \times s$
      - $W = F s \cos \theta$
    - Units
      - SI: Joule (J)
      - CGS: erg
      - 1 J = 10^7 erg
    - Types
      - Positive ($\theta = 0°$)
      - Negative ($\theta = 180°$)
      - Zero ($\theta = 90°$)
  - Energy
    - Definition
      - Capacity to perform work
    - Kinetic Energy
      - Energy due to motion
      - $K.E. = 1/2\ mv^2$
    - Potential Energy
      - Energy due to state or position
      - $P.E. = mgh$
    - Forms
      - Mechanical
      - Electrical
      - Heat
      - Light
      - Sound
      - Chemical
  - Conservation of Energy
    - Energy neither created nor destroyed
    - Total energy remains constant
    - Transformation from one type to another
  - Free Fall
    - Object falling solely under gravity
    - Total Energy (K.E. + P.E.) is constant
  - Power
    - Rate at which work is done
    - $P = W / t$
    - Unit: Watt (W)